AI Assisted Coding Lab ass-13.4

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Task 1: Refactoring Data Transformation Logic

Task Description

• Review the legacy code that computes transformed values using

an explicit loop.

• Use an AI tool to suggest a more Pythonic refactoring approach.

• Refactor the code using list comprehensions or helper functions

while preserving the output.

PROMPT:

Refactor the following Python code to make it shorter and more Pythonic using list comprehension while keeping the same output.

```
values = [2, 4, 6, 8, 10]
doubled = []
for v in values:
    doubled.append(v * 2)
print(doubled)
```

Task 2: Improving Text Processing Code Readability

• Analyze the legacy code that constructs a sentence using repeated

string concatenation.

• Ask AI to suggest a more efficient and readable approach.

• Refactor the code accordingly while k

eeping the final output

unchanged.

PROMPT:

Refactor the following Python code to improve readability and efficiency by avoiding repeated string concatenation. Keep the same output.

```python


words = ["Refactoring", "with", "AI", "improves", "quality"]
message = " ".join(words)
print(message)
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.exe
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
Refactoring with AI improves quality
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD>
```

Task 3: Safer Access to Configuration Data

- **Review the legacy code that manually checks for dictionary keys.**
- **Use AI suggestions to refactor the code using safer dictionary**

access methods.

- **Ensure the behavior remains the same for missing keys.**

PROMPT:

Refactor the following Python code to use a safer dictionary access method instead of manually checking keys. Keep the same output if the key is missing.

config = {"host": "localhost", "port": 8080}

if "timeout" in config:

print(config["timeout"])

else:

print("Default timeout used")

```python
config = {"host": "localhost", "port": 8080}
print(config.get("timeout", "Default timeout used"))
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.ex
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
Default timeout used
```

Task 4: Refactoring Conditional Logic for Scalability

Examine the multiple if–elif conditions used to determine operations.

Ask AI to suggest a cleaner, scalable alternative.

Refactor the logic using mapping techniques while preserving functionality.

PROMPT:

Refactor the following Python code to replace multiple if–elif conditions with a cleaner and scalable mapping approach while keeping the same output.

AI ASS LAB_13.4.py U

AI ASS LAB_13.4.py > action

```python
action = "divide"
x, y = 10, 2

operations = {
    "add": lambda a, b: a + b,
    "subtract": lambda a, b: a - b,
    "multiply": lambda a, b: a * b,
    "divide": lambda a, b: a / b,
}

result = operations.get(action, lambda a, b: None)(x, y)
print(result)
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.exe
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
5.0
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD>
```

Task 5: Simplifying Search Logic in Collections

Identify the explicit loop used for searching an item.

Use AI assistance to refactor the logic into a more concise and readable form.

Maintain the same output behavior.

PROMPT:

Refactor the following Python code to simplify the search logic in a collection using a more concise Python approach while keeping the same output.

```
AI ASS LAB_13.4.py U
AI ASS LAB_13.4.py > inventory
inventory = ["pen", "notebook", "eraser", "marker"]
found = "eraser" in inventory
print("Item Available" if found else "Item Not Available")
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.exe
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
Item Available
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD>
```